Inference

Ciaran Evans

Class activity

Work on the class activity, then we will discuss.

Class activity

Question 1: Is there evidence that keeping a pet bird is associated with a greater risk of lung cancer, after accounting for other lifestyle and environmental factors?

$H_0: \beta_{bird} = 0$

$H_A: \beta_{bird} > 0$

wald test: $Z = \dfrac{\hat{\beta}_{bird} - 0}{SE(\hat{\beta}_{bird})}$

Class activity

Question 1: Is there evidence that keeping a pet bird is associated with a greater risk of lung cancer, after accounting for other lifestyle and environmental factors?

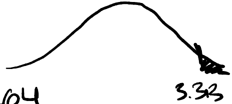
Two-Sided ($\beta = 0$ vs. $\beta \neq 0$)

```
##              Estimate Std. Error    z value    Pr(>|z
## (Intercept) -1.27063830 1.82530568 -0.6961236 0.48635145
## Bird         1.36259456 0.41127585  3.3130916 0.00092270
## SexMale     -0.56127270 0.53116056 -1.0566912 0.29065253
## SocEcLow    -0.10544761 0.46884614 -0.2249088 0.82205024
## Age         -0.03975542 0.03548022 -1.1204952 0.26250277
## SmokeYrs     0.07286848 0.02648741  2.7510612 0.00594025
## SmokeRate    0.02601689 0.02552400  1.0193110 0.30805533
```

$$z = \frac{\hat{\beta}_{bird} - 0}{SE(\hat{\beta}_{bird})} = \frac{1.3626 - 0}{0.4113} = 3.313$$

$$\text{p-value} = P(N(0,1) > 3.313)$$

$$= \frac{0.0009227}{2} = 0.000464$$



Class activity

Question 2: If an individual does keep a pet bird, is there evidence that smoking further increases their risk of lung cancer, after accounting for other factors?

Some options:

Want: $H_0: \beta_{smoke\text{-}yrs} = \beta_{smoke\,rate} = 0$

$H_A: \beta_{smoke\text{-}yrs} > 0, \ \beta_{smoke\,rate} > 0$

But this is hard!

instead:

- $H_0: \beta_{smokeyrs} = 0$ vs. $H_A: \beta_{smokeyrs} > 0$
- $H_0: \beta_{smokerate} = 0$ vs. $H_A: \beta_{smokerate} > 0$
- test both (correct for multiple comparisons w/ Bonferroni)

e.g. LRT $\longrightarrow$

- $H_0: \begin{pmatrix} \beta_{yrs} \\ \beta_{rate} \end{pmatrix} = 0$ vs. $H_A: \begin{pmatrix} \beta_{yrs} \\ \beta_{rate} \end{pmatrix} \neq 0$

Class activity

Question 2: If an individual does keep a pet bird, is there evidence that smoking further increases their risk of lung cancer, after accounting for other factors?

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## SmokeYrs     0.07286848 0.02648741  2.7510612 0.0059402
## SmokeRate    0.02601689 0.02552400  1.0193110 0.3080553
```

Class activity

Question 2: If an individual does keep a pet bird, is there evidence that smoking further increases their risk of lung cancer, after accounting for other factors?

```
m_full <- glm(Cancer ~ Bird + Sex + SocEc + Age +
                SmokeYrs + SmokeRate,
              data = birdkp, family = binomial)

m_reduced <- glm(Cancer ~ Bird + Sex + SocEc + Age,
                 data = birdkp, family = binomial)
```

$(-2 \log L_{reduced})$ $(-2 \log L_{full})$

```
m_reduced$deviance - m_full$deviance
```

```
## [1] 17.66859
```

```
pchisq(m_reduced$deviance - m_full$deviance, df = 2,
       lower.tail = F)
```

```
## [1] 0.0001456511
```

Class activity

Question 3: Consider two individuals. Individual A is male, a non-smoker, of low socioeconomic status, aged 50, who keeps a bird. Individual B is male, of low socioeconomic status, aged 50, who does not keep a bird, and who has smoked for 20 years with an average of 5 cigarettes per day. Estimate the odds ratio of lung cancer for individual A vs. individual B.

```
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```

Difference in log odds: $1.363 - 20(0.073) - 5(0.0260) = -0.227$

$OR = \exp\{-0.227\} = 0.797$

Individual A has lower odds than individual B by about 20%

Class activity

Question 4: Get a 95% confidence interval for the odds ratio.

$\hat{OR} \pm 1.96\, SE(\hat{OR})$?

challenge: getting $SE(\hat{OR})$

maybe requires delta method?

instead: get 95% CI for change in log odds, exponentiate endpoints to get CI for OR

$$\hat{\Delta} = \hat{\beta}_{bird} - 20\,\hat{\beta}_{yrs} - 5\,\hat{\beta}_{rate}$$

$$= C\hat{\beta} \qquad C = [0\ 1\ 0\ 0\ 0\ -20\ -5]$$

$$Var(C\hat{\beta}) = C\,Var(\hat{\beta})\,C^T$$

95% CI for Δ: $\hat{\Delta} \pm 1.96\sqrt{C\,Var(\hat{\beta})\,C^T}$

95% CI for OR ($= \exp\{\Delta\}$):

$$\exp\left\{\hat{\Delta} \pm 1.96\sqrt{C\,Var(\hat{\beta})\,C^T}\right\}$$

Class activity

Question 4: Get a 95% confidence interval for the odds ratio.

```
beta_hat <- coef(m_bird)
V <- vcov(m_bird)
C <- matrix(c(0, 1, 0, 0, 0, -20, -5), nrow = 1)
log_or_est <- C %*% beta_hat
log_or_sd <- sqrt(C %*% V %*% t(C))
exp(c(log_or_est - 1.96*log_or_sd,
      log_or_est + 1.96*log_or_sd))
```

```
## [1] 0.2252356 2.8317351
```